

The Scheduler's Path

Problem Statement: Implementation of a Mini Operating System Scheduler in C that simulates and compares classical CPU scheduling algorithms—First Come First Serve (FCFS), Shortest Job First (SJF), and Round Robin (RR). The system will accept user-defined inputs for arrival times, burst times, and time quantum (for RR), and will generate a clear Gantt chart to visualize execution order. Performance metrics such as waiting time, turnaround time, and response time will be calculated for comparison.

Team Name: Electrocube

Team Members:

1. **Himaneesh Singh** — Reg. No: **20254072** [Team Leader]
2. **Nikhil Saini** — Reg. No: **20254099**
3. **Shambhav Chaudhary** — Reg. No: **20254141**
4. **Srivatsan Karthikeyan** — Reg. No: **20254507**

Brief Explanation of the Problem Statement: Modern operating systems must manage multiple processes simultaneously using limited CPU resources. The challenge is determining an efficient and fair order of execution. This project aims to simulate how real OS schedulers decide which process runs at what time and how scheduling strategies affect performance. By replicating FCFS, SJF, and RR, we can observe differences in efficiency and fairness.

Objective / Why We Chose This Problem:

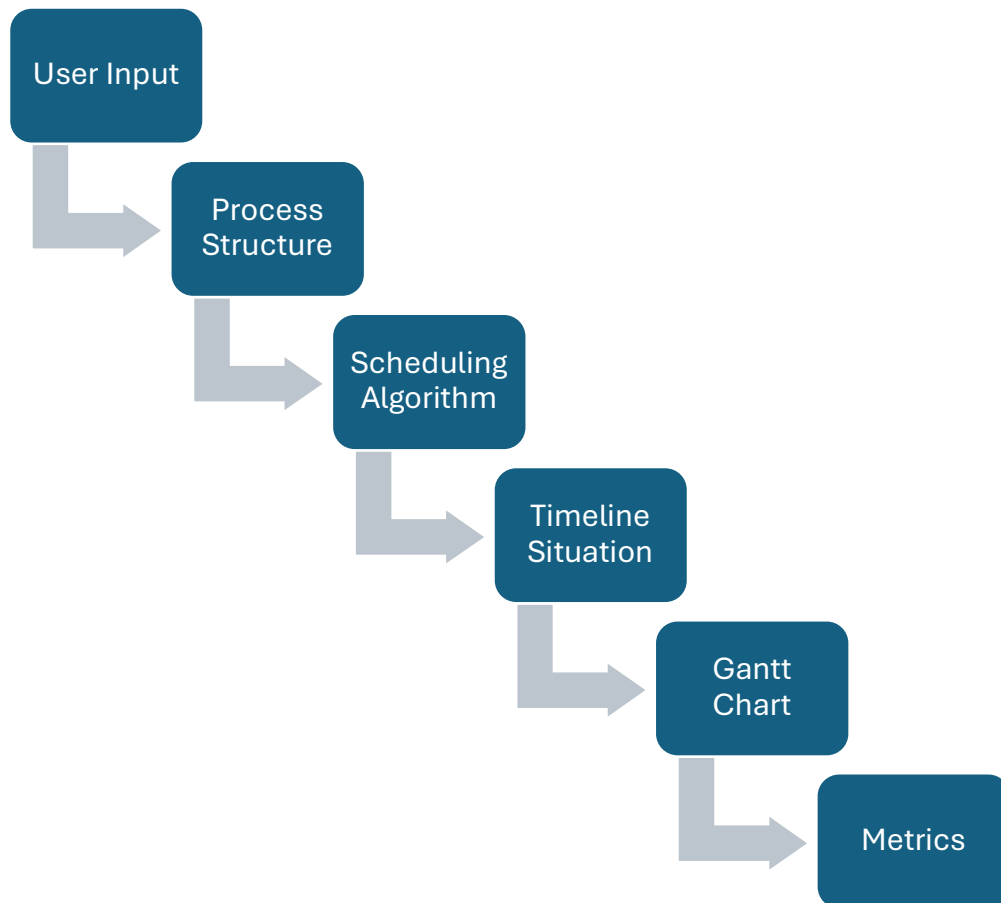
- To understand how real OSes manage multitasking
- To learn performance comparison between algorithms
- To strengthen C programming + logic skills
- To visualize CPU execution using Gantt charts
- To gain foundational experience in system-level thinking

Design/Development Approach:

1. Input Module
2. Data Structure Setup
3. Algorithm Implementation (FCFS, SJF, RR)
4. Timeline Simulation
5. Gantt Chart Generation
6. Performance Calculation
7. Time Complexity Analysis

Tools Used:

- I. C Programming Language
- II. GCC Compiler
- III. Visual Studio Code (VS Code)
- IV. Terminal/Console Output

Block/Flow Diagram:**Key Features we are going to Implement:**

- User Defined Input Process
- FCFS, SJF and RR Algorithm Implementation
- Dynamic CPU Timeline
- Queue based RR Simulation
- Gantt Chart Printing
- Waiting/Turnaround/Response Time Calculation
- Comparison of Algorithms
- Time Complexity Discussion